

IM1B AI FOR SUSTAINABILITY VIRTUAL INTERNSHIP

AI-Driven Smart Campus Energy Consumption Predictor

Sri Amaraavathi College Of Arts And Science

Rajeshwari C

Karthika A

Dharshana M

DEVELOPED IN COLLABORATION WITH IBM SKILLSBUILD & AICTE

Understanding Our Target Users

Key Target Stakeholders

Institutional Management: Seeking carbon emission cuts and budget efficiency.

Facility Administrators: Need a streamlined dashboard to allocate resources efficiently.

Campus Electricians: Require automated alerts instead of manual round-the-clock physical checks.

User Pain Points

Dynamic Timetables: Classes end early, but heavy load devices (HVAC, Projectors) remain active.

Lack of Data Insights: No centralized mechanism to predict upcoming load patterns on the local sub-grid.

The Sustainability Challenge

Campus Inefficiencies

- ⚠ Heavy electricity leakage occurs in empty seminar blocks and labs.
- ✖ Static automated timers are ineffective due to flexible hourly timetables.
- ⚙ Changing outdoor weather conditions rapidly alter cooling load requirements.

Core Framing Question

"How might we build an AI prediction engine to forecast classroom energy demands dynamically, helping system managers prevent grid waste?"

Sustainable Development Goals (SDG)



SDG 7: Affordable & Clean Energy

Optimizes base campus energy usage. High demand prediction prevents local brownouts and manages grid efficiency during extreme seasonal weather surges.



SDG 11: Sustainable Communities

Transitions our educational institution into an eco-friendly green campus model, significantly reducing localized carbon emissions and operational waste.

AI Core Solution & Logic



1. Input Parameters

Aggregates outdoor temperature, dynamic student attendance logs, and planned hour of day.



2. ML Processing

Utilizes a Supervised Linear Regression Model to map multi-variable features against historical grid metrics.



3. Actionable Output

Forecasts hourly energy loads (kWh) and issues sustainability warnings for active waste zones.

Core AI Solution & Prototype Code

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

p.random.seed(42)
records_count = 200

historical_data = {
    'Temperature_C': np.random.randint(22, 42, size=records_count),
    'Student_Count': np.random.randint(0, 180, size=records_count),
    'Hour_of_Day': np.random.randint(8, 20, size=records_count),
}

df = pd.DataFrame(historical_data)
df['Energy_Consumed_kWh'] = (df['Temperature_C'] * 1.5) + (df['Student_Count']

X = df[['Temperature_C', 'Student_Count', 'Hour_of_Day']]
y = df['Energy_Consumed_kWh']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random

ai_predictor_model = LinearRegression()
ai_predictor_model.fit(X_train, y_train)

print("="*60)
print("@  PROTOTYPE STATUS: AI CORE PREDICTIVE ENGINE ONLINE")
print("="*60)

def get_live_campus_prediction(temp, students, hour):
    input_features = pd.DataFrame([[temp, students, hour]], columns=['Temperatu
    predicted_load = ai_predictor_model.predict(input_features)[0]

    print(f"\n📡 LIVE CAMPUS MATRIX INPUTS:")
    print(f"    ↳ Ambient Temperature : {temp}°C")
    print(f"    ↳ Classroom Occupancy : {students} Students registered")
    print(f"    ↳ Planned Time Slot : {hour}:00 Hours")
    print(f"📊 ML ENGINE OUTPUT PREDICTION:")
    print(f"    ↳ Estimated Grid Demand: {predicted_load:.2f} kWh")
```

Live Code Prototype Execution

```
===== RESTART: C:/Users/rajes/OneDrive/Documents/IT
=====
🔌 PROTOTYPE STATUS: AI CORE PREDICTIVE ENGINE ONLINE
=====

📡 LIVE CAMPUS MATRIX INPUTS:
  ↳ Ambient Temperature : 34°C
  ↳ Classroom Occupancy : 120 Students registered
  ↳ Planned Time Slot   : 14:00 Hours
📊 ML ENGINE OUTPUT PREDICTION:
  ↳ Estimated Grid Demand: 92.18 kWh
✅ GRID STATUS: Balanced footprint efficiency optimized.
=====

📡 LIVE CAMPUS MATRIX INPUTS:
  ↳ Ambient Temperature : 26°C
  ↳ Classroom Occupancy : 0 Students registered
  ↳ Planned Time Slot   : 18:00 Hours
📊 ML ENGINE OUTPUT PREDICTION:
  ↳ Estimated Grid Demand: 53.94 kWh
⚠️ SUSTAINABILITY WARNING: Zero occupancy but energy spike detected. Suggesting manual HVAC cutoff overrides
=====
```

Responsible & Trustworthy AI



Data Privacy First

The system monitors headcounts via campus schedule timetables and general room registration rates. It strictly avoids individual biometric tracking, surveillance cameras, or profiling student logs.



Algorithm Explainability

We avoid complex black-box logic. By utilizing Linear Regression, the mathematical variables remain auditable and clear for any system electrician to understand and trust.

Thank You!

Driving Eco-Friendly Campus Innovation with Artificial Intelligence

20% - 30% Energy Savings

Expected institutional utility footprint reductions through smart forecasting.

Sri Amaraavathi College

Pioneering eco-tech solutions for local educational microgrids.